Spectrally Resolved Raman Lidar to Measure Backscatter Spectra of Atmospheric Three-Phase Water and Fluorescent Aerosols Simultaneously: Instrument, Methodology, and Preliminary Results

Fuchao Liu, Fan Yi, Yun He, Zhenping Yin, Yunpeng Zhang, and Changming Yu

***Abstract*— This work presents a spectrally resolved Raman lidar (SRRL) for simultaneous measurement of volume backscattering coefficient spectra (backscatter spectra for short) of atmospheric three-phase water and fluorescent aerosols. The SRRL emits 354.8-nm laser light and records both N_2 Raman echoes around 386.7 nm and spectral signals in a specially modified spectral range of ∼393–424 nm. By defining normalized spectra, a unique spectra decomposition approach is developed for successive retrieval of normalized spectra components of fluorescent aerosols, water vapor, ice water, and liquid water. Furthermore, the backscatter spectra of each component are obtained by referencing the N_2 Raman signals. Three typical measurement cases are provided. For the clear-day case, the lidar-measured normalized water vapor Raman spectra are found to be nearly invariant in shape and can serve as background spectra reference for decomposition of mixed-phase water Raman spectra. Besides, fluorescence backscatter intensities are usually weak enough to be neglected in clear areas. For the fluorescence case, the fluorescence backscatter intensities become much stronger to prevent accurate measurements of water vapor. For the mixed-phase water virga case, the decomposed backscatter spectra of condensed ice and liquid water indicate the coexistence of ice water and less amount of liquid water in the falling water virga. In conclusion, this kind of SRRL is believed to have provided the potential for simultaneous and accurate profiling of atmospheric water vapor and cloud liquid/ice water and opened up new perspectives for studies of cloud–aerosol interaction.**

***Index Terms*— Aerosol, atmospheric water, backscatter spectra, laser radar, remote sensing.**

I. Introduction

THE atmospheric water is a unique substance that actually goes through three-phase states under natural weather conditions. The water vapor is always present in the atmosphere; due to its relatively large content, it usually plays a crucial role in various atmospheric processes (e.g., radiative transfer, mesoscale circulations, and convection initiation), and knowledge of water vapor profiles is the heart of understanding these key processes [1]. Clouds are mainly composed of liquid and/or ice water. The existence of clouds essentially modulates the energy budget of the earth–atmosphere system [2] and hydrological cycle; information on the cloud microphysics and bulk properties (such as the number density, effective radius, and cloud phase and water content) is required for understanding detailed cloud impact and providing better constraints for numeric weather forecast and climate models. In summary, accurate vertical profiling of atmospheric water in all phase states with the necessary temporal and spatial resolutions is of significant importance to meet all the above demands.

The Raman lidar has already proved itself an effective tool for remote sensing of atmospheric constituents, as the Raman scattering intensity is physically proportional to the number density of a lidar-probed component. Melfi *et al.* [3] and Cooney [4] first demonstrated the technique of Raman spectroscopy for the measurement of tropospheric water vapor. A conventional water vapor Raman lidar generally utilizes a discrete detection channel to extract mainly the Q-branch backscatter signals of the vi-rotational Raman spectra of water vapor molecules with narrow-band (e.g., 0.3 nm) optical devices centering at a Raman shift of 3657 cm^{-1}. By referencing Raman signals of another permanent molecule (e.g., nitrogen), the water vapor mixing ratio (WVMR) can be subsequently retrieved [5], [6]. Thanks to the state-of-the-art optical techniques, nowadays both the ground-based (e.g., [7]–[9]) and airborne (e.g., [10]) water vapor Raman lidars are able to measure water vapor profiles from near ground to the lower stratosphere with excellent accuracy and some operate diurnally [7], [10]–[13]. The water vapor Raman lidar works well under clear weather conditions. However, it fails and might yield unreasonably large relative humidity (RH) results in cloud areas (e.g., RH > 100%) [14]. One significant reason is attributed to contamination to recorded water vapor Raman signals by Raman signals from cloud condensed water. As illustrated in Fig. 1, for an excitation laser wavelength of 354.8 nm, the bulk ice and liquid water Raman spectra [15] mainly fall into the spectral ranges of 395–407 and 396–409 nm, respectively, overlapping with the Raman

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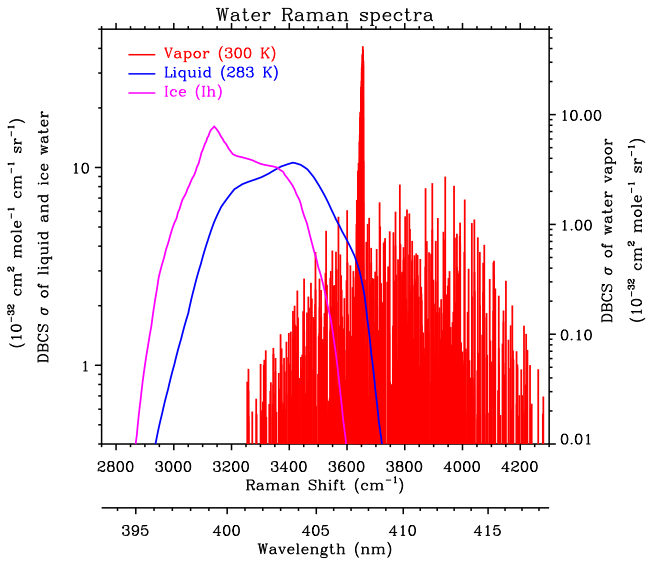


Fig. 1. Raman differential backscattering cross section σ (DBCS) of water vapor (vapor; red), liquid water (liquid; blue), and ice water (ice; magenta). The water vapor Raman spectra are theoretically calculated from Avila *et al.* [16] at a temperature of 300 K. Spectroscopic data of bulk liquid and ice (Ih) water are from Slusher and Derr [15]. Excitation laser wavelength is set to be 354.8 nm. Note the Raman spectra of three-phase water overlap in spectrum.

spectra of water vapor (401–418 nm) [16]. In the cloud area, intensified Raman backscatter signals from cloud liquid/ice water definitely enter the discrete water vapor Raman detection channel and lead to larger WVMR results. For accurate measurement of water vapor in cloud area as well as retrieval of cloud liquid/ice water content, some lidar groups [17]–[21] added additional discrete detection channel(s) in the conventional water vapor Raman lidar for simultaneous collection of Raman signals from condensed water. This kind of two (three)-channel water Raman lidar enjoys the benefits of concise lidar receiving optics and relatively larger condensed water Raman echo intensity, but suffers from the challenges for lidar system calibration [22]. Another obstacle that prevents discrete one- and two (three)-channel water Raman lidars from accurate water content measurements is aerosol fluorescence. When radiated by an ultraviolet laser light (e.g., 354.8 nm), atmospheric biogenic aerosols or aerosols containing organic materials [23]–[29] generate fluorescence which covers a wide spectral range [27], [29] and crams the entire Raman band of three-phase water. In this case, the discrete-channel water Raman lidars cannot distinguish fluorescence from water Raman signals and would often lead to false measurement results [28], [29].

According to laboratory studies [30], [31], the Raman spectra from bulk liquid water are highly dependent on temperature. For the case of liquid cloud, the Raman spectra from a swarm of quasi-spherical cloud droplets with distinct sizes in a lidar-probed volume become quite complicated due to a combined effect of structural resonances and droplet scattering phase function [17], [19], [32]–[34]. But, it is expected that the entire Raman spectral distribution of cloud liquid/ice water reflects the physical properties of clouds (e.g., cloud phase and cloud water content). In this situation, a spectrally resolved Raman lidar (SRRL) [22], [35] for the recording of Raman spectra from water vapor and liquid/ice water in clouds simultaneously is helpful for cloud studies. Moreover, an SRRL also provides possibilities for handling the Raman spectral crosstalk between mix-phased water and overcoming the contamination of water Raman signals from aerosol fluorescence [22].

Bukin *et al.* [36] reported the first measurements of atmospheric water Raman spectra from the fog in the 398.5–408.3 nm range using a scanning monochromator. Kim *et al.* [37] developed an SRRL and obtained preliminary spectral observational results from atmospheric mixed-phased water under various weather conditions [38]; the SRRL utilized a 32-channel detector for simultaneous recording of spectral signals in the 402.4–410.9 nm range. Liu and Yi [39] reported a similar SRRL for water Raman spectra measurement in the 401.7–409.0 nm range; they found that the lidar-measured water vapor Raman spectra under clear days hardly changed in shape and that the mixed-phased water Raman spectra from an aerosol/liquid layer obviously differed from those of bulk liquid water. However, all the above SRRLs only measure limited parts of the three-phase water Raman spectra and cannot directly recognize possible contaminating fluorescence from aerosols. Reichardt [22] reported an SRRL with a modified spectral detection range of 384.6–410.2 nm and presented Raman spectra measurement examples of liquid, mixed-phased, and cirrus cloud. A method using a reference spectrum between 391.6 and 394.6 nm (the gap between the Raman band of N_2 molecules and three-phase water) was proposed to serve as the "sign" of the presence of aerosol fluorescence, but the removal of fluorescence contamination to water Raman spectra was not further discussed when the "sign" indicated that fluorescence was present.

Considering all the above-mentioned contributions to water Raman lidars and challenges in accurate measurements of three-phase water, we have updated our former-version SRRL [39] to modify the lidar spectral range to be ~393–424 nm which fully covers the Raman band of three-phase water (see Fig. 1). From the lidar-measured spectra, a unique spectra decomposition method is developed for discriminating the water Raman spectra from fluorescence spectra, as well as retrieval of Raman volume backscattering coefficient spectra of water vapor, ice water, and liquid water successively. Typical measurement results by the updated SRRL are presented to verify the proposed method. It is believed that this kind of SRRL is capable of operating under various weather conditions and has provided the potential for simultaneous and accurate profiling of atmospheric three-phase water (especially condensed liquid/ice water in clouds), as well as opening up new perspectives for studies of cloud–aerosol interaction. The lidar instrument, methodology, measurement examples, and summary and outlook are presented subsequently in the following sections.

II. Lidar Instrument

Currently, the updated SRRL is located at our Atmospheric Observatory on the campus of Wuhan University (30.5°N,

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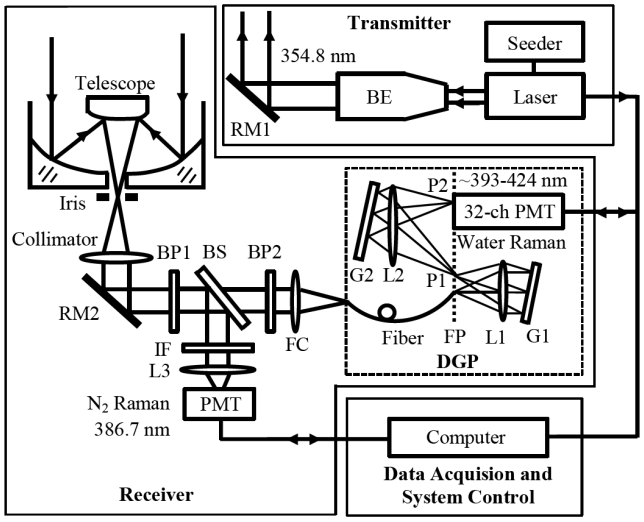


Fig. 2. Schematic layout of the spectrally resolved Raman lidar (SRRL) system. BE, beam expander; RM, reflecting mirror; BP, bandpass filter; BS, beam splitter; L, lens; FC, fiber coupler; G, grating; IF, interference filter; PMT, photomultiplier tube; DGP, dual-grating polychromator; FP, focal plane; P, position of focused light spot.

114.4°E, 72 m above sea level) where a series of lidar systems [40]–[47], including a 532-nm polarization lidar [43], had been developed. Compared to the former-version SRRL [39], the updated SRRL has an extended spectral detection range of ~393–424 nm for special considerations in dealing with aerosol fluorescence issues and recording the full Raman spectra of all three-phase water. Consequently, the lidar receiving optics of the updated SRRL has been redesigned and manufactured. Fig. 2 provides a schematic layout of the updated SRRL system.

The lidar transmitter employs a tripled injection-seeded Nd:YAG laser to generate an emission of ~100 mJ per pulse at 354.8 nm with a repetition rate of 20 Hz, a pulsewidth of 6 ns, and linewidth of <0.1 pm. A subsequent 5× beam expander (BE) is equipped to reduce the divergence of the output laser beam before it is guided into the atmosphere zenithward by a reflecting mirror (RM1). The RM1 has a reflectivity of >99.5% for the incident laser light and is positioned on a 2-D electronically adjustable mount. At the receiver side, a 450-mm Cassegrain telescope with a focal length of 4 m collects the atmospheric backscatter. After passing through an iris, the collected backscatter lights are first collimated by a collimator (focal length = 120 mm), then guided to be horizontal by another reflecting mirror (RM2). For restriction of strong elastic echo signals, a first bandpass filter (BP1) is placed after RM2. The BP1 (Semrock, USA) has average transmission of >94% in the 387.0–447.0 nm range, but rejects out-band lights (including elastic lights around 354.8 nm) with a magnitude of >6 orders. After BP1, the filtered lights are divided into two parts by a beam splitter (BS): 10% is reflected to enter an interference filter (IF), while the remaining ~90% is transmitted to enter a second bandpass filter (BP2). The IF (Andover, USA) has a full-width at half-maximum (FWHM) of 0.3 nm centered at 386.7 nm, picking out mainly the Q-branch signals of the N_2 vi-rotational Raman spectra. It is of a peak transmission of ~30% around 386.7 nm and a suppression of six orders of magnitude around 354.8 nm. The 386.7-nm light is then focused onto the photosensitive surface of a photomultiplier tube (PMT; R4632, Hamamatsu, Japan) by a collimating lens (L3) and recorded as a reference signal. The PMT has a quantum efficiency of ~25% around 386.7 nm. BP2 (Barr, USA) is a customized bandpass filter. It has a transmission of >80% to the signals in the ~393–424 nm range and suppression of >3 orders of magnitude to out-band signals (including the elastic signal around 354.7 nm and the O_2 and N_2 vi-rotational Raman signals around 375.4 and 386.7 nm, respectively). The signal lights exiting the BP2 are coupled into a dual-grating polychromator (DGP) by a fiber coupler (FC) with a coupling efficiency of >70%. The FC has a focal length of 50 mm, a clear aperture of 22 mm, and a numerical aperture (NA) of 0.23.

The DGP is specially designed and manufactured for the updated SRRL system. It comprises a multi-mode optical fiber, two separate single-grating polychromators, and a 32-channel detector. Each single-grating polychromator comprises both a lens (L) and a grating (G) which are quasi-Littrow configured. The two single-grating polychromators have optical axes parallel to each other in the same horizontal plane with a distance of 53.69 mm and share the same vertical focal plane (FP). The fiber transmits the coupled lights into the DGP. The surface center of the fiber output locates at the focal point of the L1. So that for transmitted lights with a given wavelength of λ in the ~393–424 nm range, they become parallel after passing through the L1. Then the parallel lights are diffracted by G1 and refocused by L1 on the FP of L1 (the position of the focused light spot is marked by P1 as shown in Fig. 2). Since P1 is on the FP of L2, the lights from P1 with a wavelength of λ again become parallel after passing through L2, then be diffracted by G2 and refocused by L2 on the FP (the position of the focused light spot is marked by P2 as shown in Fig. 2). For lights with distinct wavelengths of λ in the ~393–424 nm range, they have finally dispersed apart and focused at different positions on the FP of L2. Note the first single-grating polychromator performs an initial spectral dispersion of the incoming signals in the ~393–424 nm range with a linear dispersion of 0.18 mm nm^{-1}, while the second single-grating polychromator further disperses the spectral signals with an overall linear dispersion of 1.0 mm nm^{-1}. The 32-channel detector is positioned exactly on the FP of the second single-grating polychromator, registering the dispersed spectra in the ~393–424 nm range with a 1.0-nm spectral resolution.

The fiber is customized (Fiberguide, USA) to have a core diameter of 0.6 mm and an NA of 0.12. The telescope field of view is confined to be ~0.3 mrad by the fiber. L1 has a clear aperture of 100 mm and a focal length of 300 mm, while L2 has a clear aperture of 100 mm and a focal length of 400 mm. Both L1 and L2 are coated to guarantee transmission of >99% in the ~393–424 nm range. The first grating (Newport, USA) is a planar reflection blazing grating with a groove density of 600 gr mm^{-1} and a blazing angle of 6.89°. It operates with a working angle of 9.27°, diffraction order of 1, and diffraction efficiency of >75%. The second

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grating is similar to the first grating in type, but has a blazing angle of 21.10°. It operates with a working angle of 21.72°, diffraction order of 3, and diffraction efficiency of >70%. The 32-channel detector (Licel, DE) mainly contains a 32-channel linear-array PMTs (32-ch PMT; H7260, Hamamatsu, Japan). Each photosensitive surface of the individual PMT has a geometrical size of 0.8-mm width and 7.0-mm length. The spacing distance and dead area interval between two adjacent PMTs are 1.0 and 0.2 mm, respectively. The photosensitive surfaces of the 32 PMTs are located exactly on the FP of L2, and their 7.0-mm sides are adjusted to be parallel to the vertical direction. The 32 PMTs have quantum efficiency of >40% in the ∼393–424 nm range.

The DGP has a suppression of >6 orders of magnitude to the elastic signal around 354.8 nm, as well as a suppression of >3 orders of magnitude, respectively, to the O_2 and N_2 vi-rotational Raman signals around 375.4 and 386.7 nm. Note the joint usage of the DGP and bandpass filters (BP1 and BP2) yields a suppression of more than 15 orders of magnitude to the elastic signal around 354.8 nm, while a combination of the BP2 and DGP results in a suppression of more than six orders of magnitude to the O_2 and N_2 vi-rotational Raman signals around 375.4 and 386.7 nm, respectively. Such suppressions ensure that the desired weak water Raman and possible aerosol fluorescence signals in the spectral range of ∼393–424 nm are not contaminated by the extremely strong elastic signal as well as the strong O_2 and N_2 vi-rotational Raman signals.

The data acquisition and system control unit mainly contains a computer and a time-sequence circuit. An MCS-pci card (Ametek, USA) acquires the N_2 Raman signal with a bin width of 200 ns (corresponding to an altitude resolution of 30 m), while a Licel Multispectral Lidar Detector (Licel, DE) records the light signals from the 32-ch PMTs with the same bin width of 200 ns. The photon counts (PCs) from 11 000 laser shots are accumulated to produce one original PC profile at each channel, yielding a time resolution of ∼10 min for the raw data (50 s left for data transfer). The whole system is synchronized by a DG645 (Stanford Research System, USA). The computer is utilized for overall data acquisition, system control, and real-time monitoring of the lidar operation status. Table I summarizes the main system specifications for the overall SRRL.

TABLE I
MAIN SYSTEM SPECIFICATIONS FOR THE SRRL

Transmitter	
Laser	Nd:YAG, ~100 mJ/Pulse, 20 Hz; linewidth<0.1 pm@354.8 nm;
BE	5×, RE>99.5%;
Receiver	
Telescope	D=450 mm, FL=4 m, FOV=~0.3 mrad
Filters	BP1: TR>94%@387-447 nm, OOD>6; BP2: TR>80%@393-424 nm, OOD>3; IF: CWL=386.7 nm, BW=0.3 nm; Tp>30%, OOD>6;
Beam splitter	RE=10%, TR=~90%;
Fiber coupler	FL=50 mm, CA=22 mm, NA=0.23, CE>70%;
DGP	OLD: 1.0 mm nm^{-1}@393-424 nm, OOD>3; Fiber: D=0.6 mm, NA=0.12; L1: CA=0.1 m, FL=0.3 m, TR>99%; L2: CA=0.1 m, FL=0.4 m, TR>99%; G1: 600 gr mm^{-1}, BA=6.89º ,WA=9.27º; DO=1, DE>75%; G2: 600 gr mm^{-1}, BA=21.10º ,WA=21.72º; DO= 3, DE>70%;
Data digitizing	
N_2 channel	PMT: R4632, QE=25%@386.7 nm; MCS-pci card: Bin width=200 ns;
Spectral channels	32-ch PMT: H7260, QE>40%; Licel Multispectral detector: Bin width=200 ns;

FL: focal length; TR: transmittance; OOD: outband optical density; CWL: central wavelength; BW: bandwidth; Tp: peak transmittance; RE: reflectivity; CE: coupling efficiency; OLD: overall linear dispersion; BA, blazing angle; WA, working angle; DO, diffraction order; DE: diffraction efficiency; QE: quantum efficiency.
All TR and DE data are provided by manufacturer.

III. METHODOLOGY

A. System Test and Calibration

After reassembly of the SRRL receiving optics, the DGP is first tested and calibrated. The testing procedure makes use of the two spectral lines at 404.66 and 407.78 nm emitted by a low-pressure mercury lamp; for details, refer to our former work [39]. An area-array CCD (S7031-1008S, Hamamatsu) was placed instead of the 32-ch PMT on the FP of L2. The CCD has an active image area of 1024 × 250 pixels with each side length of 24 μm. Then the light from a low-pressure Hg lamp was fed into the DGP via the fiber. After dispersion by the DGP, the CCD registered separate Hg spectral lines at 404.66 and 407.78 nm with geometric FWHM of ∼0.6 mm and a peak-to-peak spacing of 3.096 mm. This confirms that the DGP has a linear dispersion of ∼0.99 mm nm^{-1}. In the calibrating procedure, a commercial double-grating (2400 gr mm^{-1}) monochromator (M833, Belarus) together with a white light source (HPLS-30, Thorlabs) were used to yield a wavelength-adjustable monochromatic light [42]. The entrance and exit slits of the monochromator were both set to be 20 μm, so that the output monochromatic light had an FWHM of ∼0.02 nm around 407.6 nm and was focused onto the photosensitive surface of the 16th channel of the 32-ch PMT. Then through a clear-day observational experiment, G2 was finely adjusted so that the 16th channel of the 32-ch PMT recorded the strongest signals from the Q-branch Raman spectra of atmospheric water vapor (around 407.66 nm). At this time, G2 was fixed without any further adjustment. Based on this adjusted configuration, the central wavelength for the kth detection channel could be calculated simply by the following expression:

$$\lambda_k = (k - 16) + 407.66, \quad k = 1, 2, \ldots, 32. \tag{1}$$

The central wavelength λ_k for each detection channel was further verified by a scanning experiment [42]: the white light source and commercial monochromator were first set to output quasi-monochromatic light centered at λ_k defined by (1) into the DGP; then it was found that the DGP precisely dispersed the quasi-monochromatic light centered at λ_k onto the kth detection channel.

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B. Lidar-Measured Normalized Spectra

The lidar equation for each spectral detection channel can be expressed in the following form (background N^{B} subtracted):

$$\begin{cases} N_k(z) = N_k^{\mathrm{T}} - N_k^{\mathrm{B}} = \dfrac{C_k \cdot O_k(z)}{z^2} \cdot \beta_k^{\mathrm{T}}(z) \cdot \tau(\lambda_0) \cdot \tau(\lambda_k) \\ \beta_k^{\mathrm{T}} = \beta_k^{\mathrm{F}} + \beta_k^{\mathrm{V}} + \beta_k^{\mathrm{L}} + \beta_k^{\mathrm{I}} \end{cases} \tag{2}$$

where $N_k(z)$ stands for the background-subtracted PCs at height z for the kth detection channel. Background N^{B} is determined from the PCs at the far end of the lidar detection range. C is the system constant, O is the overlap factor, and τ the atmospheric transmission at laser wavelength λ_0 or backscattered signal wavelength λ_k. The total (T) volume backscattering coefficient β^{T} can be contributed by fluorescence (F) from aerosols and Raman signals from water vapor (V), liquid water (L), and ice water (I).

Usually, the 16th channel has the strongest signal intensity as it mainly extracts the Q-branch Raman signal from atmospheric water vapor. Here, referring to the idea proposed by Reichardt [22], we define original normalized spectra $SO^{\mathrm{T}} = \{SO_k^{\mathrm{T}}\}$ in an aggregate form and each term can be expressed as

$$SO_k^{\mathrm{T}} = \frac{N_k}{N_{16}} = \frac{N_k^{\mathrm{T}} - N_k^{\mathrm{B}}}{N_{16}^{\mathrm{T}} - N_{16}^{\mathrm{B}}} = \frac{C_k \cdot O_k \cdot \tau(\lambda_k)}{C_{16} \cdot O_{16} \cdot \tau(\lambda_{16})} \cdot \frac{\beta_k^{\mathrm{T}}}{\beta_{16}^{\mathrm{T}}}. \tag{3}$$

Considering that the wavelength difference between λ_k and λ_{16} is small (less than 17 nm), the difference for corresponding atmospheric back-trip transmission can be ignored: $\tau(\lambda_k) \approx \tau(\lambda_{16})$. Besides, the lidar receiver utilizes fiber optics to transmit the spectral signals, so that the assumption of $O_k \approx O_{16}$ can be reasonably valid. A calibration experiment similar to [42] was made and the ratio values of each C_k to C_{16} were consequently determined. Then, we get total normalized spectra S^{T}

$$S_k^{\mathrm{T}} = SO_k^{\mathrm{T}} \cdot \frac{C_{16}}{C_k} \approx \frac{\beta_k^{\mathrm{T}}}{\beta_{16}^{\mathrm{T}}} = \frac{\beta_k^{\mathrm{F}} + \beta_k^{\mathrm{V}} + \beta_k^{\mathrm{L}} + \beta_k^{\mathrm{I}}}{\beta_{16}^{\mathrm{T}}}. \tag{4}$$

From (4), we can define the normalized spectra component of fluorescence S^{F}

$$S^{\mathrm{F}} = \left\{S_k^{\mathrm{F}}\right\} \approx \left\{\beta_k^{\mathrm{F}}/\beta_{16}^{\mathrm{T}}\right\}. \tag{5}$$

Then normalized Raman spectra components of water vapor S^{V}, liquid water S^{L}, and ice water S^{I} can be similarly defined. The advantage of defining normalized spectra here, as shall be seen later, is that the spectra decomposition process for retrieving spectra components of each-phase water from the total normalized water Raman spectra makes use of overall Raman spectra shape of each-phase water rather than water content (water Raman signal intensity).

C. Retrieval of Normalized Fluorescence Spectra Component

According to laboratory measurement results of bulk liquid/ice water Raman spectra and theoretical calculations on water vapor Raman spectra (see Fig. 1), the Raman band of water vapor, liquid water, and ice water mainly fall into 401–418 (k = 9–26), 396–409 (k = 4–17), and 395–407 (k = 3–15) nm ranges, respectively, for a radiation laser wavelength of 354.8 nm. Hence, the Raman band of three-phase water takes the spectral range of 395–418 nm (k = 3–26). While spectral signals are in the 393–394 (k = 1–2) and 419–424 (k = 27–32) nm ranges, they are mostly contributed by aerosol fluorescence. So, spectral signals in both the 393–394 and 419–424 nm ranges can be regarded as the "sign" of aerosol fluorescence. The aerosol fluorescence usually spans a large spectral range and varies relatively smoothly in a small spectral domain, and hence a linear fitting is applied to approximate the whole normalized fluorescence spectra component S^{F} in the limited lidar spectral range

$$\begin{cases} S_k^{\mathrm{F}} \approx \beta_k^{\mathrm{F}}/\beta_{16}^{\mathrm{T}}, \quad k = [1,2] \cup [27,28,\ldots,32] \\ \Downarrow \text{ linear fitting} \\ \left\{S_k^{\mathrm{F}}\right\}, \quad k = 1,2,\ldots,32. \end{cases} \tag{6}$$

Note here an assumption that the fluorescence spectra vary linearly in the lidar spectral range is made. Even if this assumption is not rigorous, it provides a measure for estimating fluorescence spectra in the 395–418 nm range. Actually, this estimation works as later measurement examples demonstrate it.

D. Retrieval of Normalized Three-Phase Water Raman Spectra Components

Subtracting the fitted normalized fluorescence spectra component S^{F} from the total normalized spectra S^{T} yields normalized Raman spectra components of three-phase water S^{W}

$$S_k^{\mathrm{W}} = S_k^{\mathrm{V}} + S_k^{\mathrm{L}} + S_k^{\mathrm{I}} = S_k^{\mathrm{T}} - S_k^{\mathrm{F}}, \quad k = 3,4,\ldots,26. \tag{7}$$

Under fine weather conditions or if we choose clear areas without clouds and with very little aerosols, the lidar-measured water Raman spectra shall be mainly contributed by water vapor. Then we get normalized water vapor Raman spectra component S^{V}

$$S_k^{\mathrm{V}} = S_k^{\mathrm{W}} = S_k^{\mathrm{T}} - S_k^{\mathrm{F}}, \quad k = 9,10,\ldots,26. \tag{8}$$

Note that S^{V} are convenient and easy to be measured by the SRRL as water vapor is always present in the atmosphere. According to our earlier work [39], the overall shape of normalized water vapor Raman spectra is nearly invariant. This feature implies that the lidar-measured S^{V} can serve as the "background spectra reference" for the decomposition of three-phase water Raman spectra. Specifically, the condensed water Raman spectra range from 395 to 409 nm in theory (see Fig. 1), so that the water Raman spectra in the 410–418 nm range can only be contributed by water vapor. Thus, for general cases, by subtracting S^{V} from S^{W}, we obtain normalized condensed liquid/ice water Raman spectra component $S^{\mathrm{C}} = S^{\mathrm{L,I}}$

$$\begin{cases} S_k^{\mathrm{L,I}} = S_k^{\mathrm{W}} - S_k^{\mathrm{V}}, & k = 3,4,\ldots,17 \\ S_k^{\mathrm{W}} \equiv S_k^{\mathrm{V}}, & k = 18,20,\ldots,26. \end{cases} \tag{9}$$

In the case of high-level cirrus, ice virga, or snow, when the condensed water can be regarded to be composed of only

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(or mostly) ice water, we can get normalized ice water Raman spectra component S^{I} by subtracting S^{V} from S^{W}

$$S_k^{\mathrm{I}} = S_k^{\mathrm{W}} - S_k^{\mathrm{V}}, \quad k = 3, 4, \ldots, 15. \tag{10}$$

Considering that the hydrogen bonds formed between H_2O molecules in ice water are relatively stable compared to those in liquid water, the overall shape of normalized ice water Raman spectra has weaker dependence on temperature [50]. Like normalized water vapor Raman spectra, the normalized ice water Raman spectra can also serve as background spectra reference for decomposition of the condensed water Raman spectra

$$\begin{cases} S_k^{\mathrm{L}} = S_k^{\mathrm{L,I}} - S_k^{\mathrm{I}}, & k = 4, 5, \ldots, 17 \\ S_3^{\mathrm{L,I}} \equiv S_3^{\mathrm{I}}. \end{cases} \tag{11}$$

Note that the liquid water Raman spectra are highly dependent on temperature according to previous laboratory studies [48]–[50], so the overall shape of liquid water Raman spectra shall be variant, indicating that S^{L} is not suitable to serve as background spectra reference and are the last to be obtained in the spectral decomposition process.

So far from the lidar-measured total normalized spectra S^{T}, the above-introduced spectral decomposition method yields successively the normalized spectral component of aerosol fluorescence S^{F} and normalized Raman spectra components of water vapor S^{V}, ice water S^{I}, and liquid water S^{L}. The key points for applying the method are: 1) the fluorescence spectra are assumed to vary smoothly in the limited lidar spectral range; 2) the normalized Raman spectra of water vapor are measured in advance by the SRRL and serve as background spectra reference for the decomposition of three-phase water Raman spectra; and 3) the normalized Raman spectra of ice water are measured by the SRRL in advance and serve as background spectra reference for the decomposition of condensed water Raman spectra. Considering ice water is not always present in the atmosphere, an alternative choice is utilizing spectral data measured in the lab by Slusher and Derr [15] for simulating the normalized Raman spectra of ice water, but this is to be checked by the lidar-measured normalized ice water Raman spectra.

E. Retrieval of Volume Backscattering Coefficient Spectra

After retrieval of each normalized spectral component, the PC in each detection channel contributed by both fluorescent aerosols and three-phase water can be easily obtained from (3) and (4)

$$\begin{cases} N_k^{\mathrm{F}} = S_k^{\mathrm{F}} \cdot N_{16}/(C_{16}/C_k), & k = 1, 2, \ldots, 32 \\ N_k^{\mathrm{V}} = S_k^{\mathrm{V}} \cdot N_{16}/(C_{16}/C_k), & k = 9, 10, \ldots, 26 \\ N_k^{\mathrm{L}} = S_k^{\mathrm{L}} \cdot N_{16}/(C_{16}/C_k), & k = 4, 5, \ldots, 17 \\ N_k^{\mathrm{I}} = S_k^{\mathrm{I}} \cdot N_{16}/(C_{16}/C_k), & k = 3, 4, \ldots, 15. \end{cases} \tag{12}$$

In particular, the 16th detection channel extracts mainly the Q-branch Raman signal from water vapor

$$N_{16}^{\mathrm{V}} = S_{16}^{\mathrm{V}} \cdot N_{16}. \tag{13}$$

Combining N_{16}^{V} and the referencing N_2 Raman channel signal N_{N2}, the WVMR q (defined as the mass ratio of water vapor to dry air in a given volume) can be obtained following the method utilized by conventional water vapor Raman lidars [51], [52]:

$$q(z) = C_q \cdot \Delta\tau \cdot \frac{N_{16}^{\mathrm{V}}}{N_{\mathrm{N2}}} \tag{14}$$

where C_q is a system constant that can be calibrated using accompanying radiosonde WVMR data. $\Delta\tau$ is the atmospheric back-trip transmission difference due to distinct N_2 and water vapor Raman signal wavelengths; this transmission difference is usually very small [51], [52] and is neglected here.

Then, we can derive the Raman volume backscattering coefficient of water vapor β_{16}^{V}

$$\left.\begin{array}{l} \beta_{16}^{\mathrm{V}} = n^{\mathrm{V}}(z) \cdot \sigma^{\mathrm{V}} \\ q(z) = \frac{m_{\mathrm{V}}}{m_{\mathrm{air}}} = \frac{18}{28.6} \cdot \frac{n^{\mathrm{V}}(z)}{n^{\mathrm{air}}(z)} \end{array}\right\} \Rightarrow \beta_{16}^{\mathrm{V}} = \frac{28.6}{18} \cdot \sigma^{\mathrm{V}} \cdot q(z) \cdot n^{\mathrm{air}}(z) \tag{15}$$

in which n is the molecular number density and m means the mass of water vapor (V) or dry air (air). σ^{V} is the effective Raman differential backscattering cross section of water vapor and its value can be set to be 6.85×10^{-30} cm^2 sr^{-1} mole^{-1} [16] for a 354.8-nm laser wavelength and a 0.8-nm bandwidth for the 16th detection channel. The number density of dry air $n^{\mathrm{air}}(z)$ can be inferred from concurrent radiosonde data.

From (4) and (15), the total volume backscattering coefficient β_{16}^{T} at λ_{16} can be retrieved as

$$\beta_{16}^{\mathrm{T}} = \beta_{16}^{\mathrm{V}}/S_{16}^{\mathrm{V}}. \tag{16}$$

Then combing (4) and (16), the lidar-measured volume backscattering coefficient spectra of aerosol fluorescence $\{\beta_{\mathrm{k}}^{\mathrm{F}}\}$ and Raman volume backscattering coefficient spectra of water vapor $\{\beta_{\mathrm{k}}^{\mathrm{V}}\}$, liquid water $\{\beta_{\mathrm{k}}^{\mathrm{L}}\}$, and ice water $\{\beta_{\mathrm{k}}^{\mathrm{I}}\}$ can be determined as

$$\begin{cases} \beta_k^{\mathrm{F}} = S_k^{\mathrm{F}} \cdot \beta_{16}^{\mathrm{T}}, & k = 1, 2, \ldots, 32 \\ \beta_k^{\mathrm{V}} = S_k^{\mathrm{V}} \cdot \beta_{16}^{\mathrm{T}}, & k = 9, 10, \ldots, 26 \\ \beta_k^{\mathrm{L}} = S_k^{\mathrm{L}} \cdot \beta_{16}^{\mathrm{T}}, & k = 4, 5, \ldots, 17 \\ \beta_k^{\mathrm{I}} = S_k^{\mathrm{I}} \cdot \beta_{16}^{\mathrm{T}}, & k = 3, 4, \ldots, 15. \end{cases} \tag{17}$$

For short, hereafter, we use backscatter spectra instead of volume backscattering coefficient spectra in this work.

F. Retrieval of Integrated Total Volume Backscattering Coefficient

From the above-retrieved backscatter spectra, the integrated total volume backscattering coefficient β_Λ for each component can be consequently obtained

$$\beta_\Lambda^{\mathrm{F,V,I,L}} = \int \beta^{\mathrm{F,V,I,L}} \cdot d\lambda \approx \sum_k \beta_k^{\mathrm{F,V,I,L}} \cdot \Delta\lambda. \tag{18}$$

Here, $\Delta\lambda$ (=0.8 nm) is the spectral width for each of the 32 spectral detection channels.

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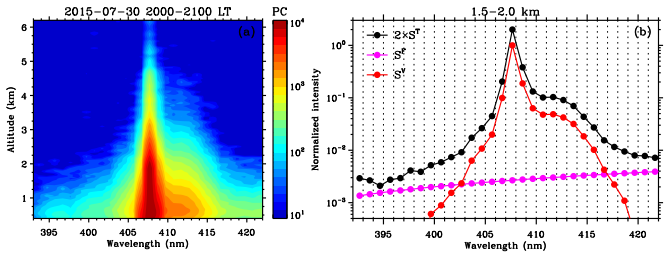


Fig. 3. (a) Altitude-resolved spectral signals (photon counts, PC) measured by the SRRL during 2000–2100 LT on July 30, 2015. (b) Total normalized spectra S^{T} derived by the lidar (black) and decomposed normalized spectral components of fluorescence spectra S^{F} (magenta) and water vapor Raman spectra S^{V} (red). Photons accumulated in 1.5–2.0 km altitude range are used to generate the normalized spectra. Note the plotted S^{T} is multiplied by a factor of 2 for better discrimination.

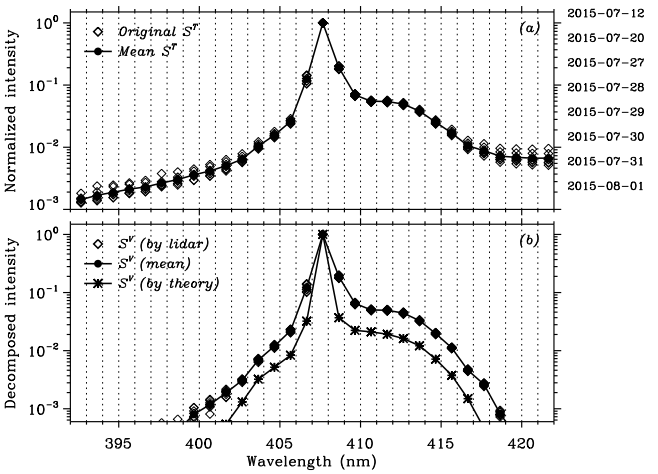


Fig. 4. (a) Lidar-measured nightly-mean total normalized spectra S^{T} on eight separate clear nights (diamond) and their total mean (solid dot). The specific dates for the eight nights are shown on the right most of the panel. (b) Retrieved normalized water vapor Raman spectra S^{V} for the eight nights (diamond) and their total mean (solid dot). For comparison, the theoretical water vapor Raman spectra [16] are simulated with a resolution of 1.0 nm and a bandwidth of 0.8 nm and added (asterisk). Note the lidar-retrieved S^{V} are nearly invariant and quite similar in shape to the theoretically simulated spectra, but broadened by the lidar receiving optics.

According to the definition of water Raman volume backscattering coefficient, (18) can also be expressed as

$$\beta_{\Lambda}^{\mathrm{V,I,L}} = n^{\mathrm{V,I,L}} \cdot \int \sigma^{\mathrm{V,I,L}} \cdot d\lambda = n^{\mathrm{V,I,L}} \cdot \sigma_{\Lambda}^{\mathrm{V,I,L}} = \eta \cdot \sigma_{\Lambda,\mathrm{bulk}}^{\mathrm{V,I,L}} \cdot n^{\mathrm{V,I,L}} \tag{19}$$

where $\sigma_{\Lambda}^{\mathrm{V,I,L}}$ stands for integrated total Raman differential backscattering cross section of atmospheric water vapor or liquid/ice water. For $\sigma_{\Lambda}^{\mathrm{I,L}}$, an enhancement factor η [33] can be introduced to represent the structural resonance effect of cloud particles and relate $\sigma_{\Lambda}^{\mathrm{I,L}}$ with integrated total Raman differential backscattering cross section of bulk ice/liquid water $\sigma_{\Lambda,\mathrm{bulk}}^{\mathrm{I,L}}$. Note $\sigma_{\Lambda,\mathrm{bulk}}^{\mathrm{I,L}}$ is a known constant as it has already been determined by laboratory experiments [50]. Finally, the integrated total volume backscattering coefficient $\beta_{\Lambda}^{\mathrm{V,I,L}}$ actually provides a measure of water molecular number density $n^{\mathrm{V,I,L}}$ (water content) in the lidar-probed volume.

IV. Measurement Examples

For the protection of the lidar system, currently the SRRL mainly operates at night under visibly fine weather conditions, or sometimes with light aerosols/clouds present, but without precipitation. In this section, three typical kinds of measurement results (namely clear-day case, fluorescence case, and mixed-phase water virga case) are presented to verify the proposed method and test the lidar performance.

A. Clear-Day Measurement Case

Fig. 3 provides a measurement example performed during 2000–2100 local time (LT) on 30 July 2015 when the atmosphere was clear. Fig. 3(a) plots the altitude-resolved spectral signals (PC) recorded by the SRRL. As seen in Fig. 3(a), the spectral signals exhibited a prominent peak at the 16th channel, which was contributed mainly by the Q-branch Raman signals of water vapor around 407.66 nm. The water vapor Q-branch Raman signals were strong with an altitude extension of ~6 km, while spectral signals on both sides were relatively weak and mainly confined below ~2 km. Besides, at an altitude below 2 km, the intensities of spectral signals on each side of the prominent peak generally depicted a clear-cut decreasing trend. Using PC data accumulated in an altitude range of 1.5–2.0 km, Fig. 3(b) plots the corresponding normalized total spectra S^{T} (black; multiplied with a factor of 2 for better discrimination). Similarly, as shown in Fig. 3(b), S^{T} peaked at the 16th channel, indicating the presence of water vapor in the selected altitude range, while on each side of this peak, S^{T} generally had smaller magnitudes with farther spectral distance off the peak. In detail, the magnitudes of S^{T} were less than 0.0015 in the spectral range of ~393–394 nm, while less than 0.0069 in the ~420–424 nm range. Using S^{T} in these two spectral ranges, the normalized spectral component of fluorescence S^{F} was first fitted (magenta). Then by subtracting S^{F} from S^{T}, we got a normalized Raman spectral component of water vapor S^{V} (red), as only water vapor (rather than condensed water) could be present in the clear atmosphere.

As stated earlier, the decomposition of mixed-phase water Raman spectra needs lidar-retrieved normalized water vapor Raman spectra in advance to serve as "background spectra reference"; besides, the overall spectral shape of lidar-retrieved normalized water vapor Raman spectra shall be nearly invariant. Considering this issue, S^{V} is further investigated based on lidar measurements of eight separate clear nights (see Fig. 4). Fig. 4(a) over-plots the nightly-mean total normalized spectra S^{T} (diamond). The corresponding observation dates are provided on the right most of the panel. Note that, for each observational night, we first got S^{T} with 1-h integration periods in the 1.0–1.5 km altitude range, then by averaging of 9–10 groups of S^{T} we obtained nightly-mean S^{T}. Finally, using the eight groups of nightly-mean S^{T}, we got their total mean S^{T} (solid dot). As seen in Fig. 4(a), the lidar-measured nightly-mean S^{T} were generally stable near the peak, but showed fluctuations at both spectral sides far away from the peak. For a very clear atmosphere without clouds and

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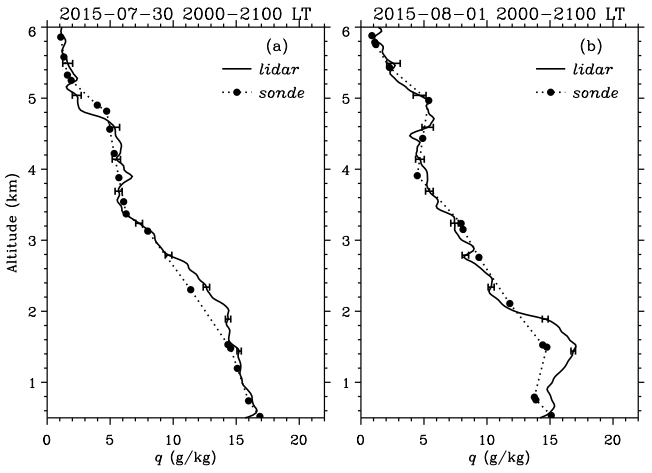


Fig. 5. Lidar-measured profiles of water vapor mixing ratio q (solid line) on the dates of: (a) July 30, 2015 and (b) August 1, 2015. Photons between 2000 and 2100 LT were accumulated to generate the results with a vertical resolution of 30 m. Error bars stand for 1-σ uncertainties. Concurrent radiosonde data (solid circle) are provided for comparison.

with very little aerosols, the intensities of spectral signals at both spectral sides were very weak with a low signal-to-noise ratio (SNR), leading to observable fluctuations. After the spectral decomposition, Fig. 4(b) plots the lidar-retrieved S^{V} for the eight separate nights (diamond) and their total mean (solid dot). It is clearly indicated in Fig. 4(b) that the lidar-retrieved S^{V} were nearly invariant in the spectral range of 400–419 nm. For comparison, the theoretical Raman spectra of water vapor [16] were simulated with a resolution of 1.0 nm and a bandwidth of 0.8 nm (asterisk). It is found that the lidar-retrieved total mean S^{V} was quite similar in shape to the theoretically simulated spectra, but broadened by the lidar receiving optics. This feature was also verified for other altitude ranges (e.g., 1.5–2.0 km) and observational dates, coinciding with our former study [39]. Note the atmospheric temperatures in the 1.0–2.0 km altitude range for these eight nights ranged from 287 to 299 K. Until now, we believe that we have obtained lidar-measured S^{V} and hereafter utilize the total mean S^{V} as "background spectra reference" for applying the spectral decomposition.

Since the WVMR q serves as a prerequisite for inferring the subsequent backscatter spectra, Fig. 5 presents examples of the q profiles measured by the SRRL (solid line) with a vertical resolution of 30 m. For generating the q profiles, PC during 2000–2100 LT are accumulated. Besides, concurrent radiosonde data (solid circle) are also provided for comparison. The radiosondes were released at 2000 LT from the Wuhan Weather Station (~23.4 km northwest of our lidar site). It is seen in Fig. 5 that the lidar-derived WVMR results generally coincide with those from the radiosonde. The observed discrepancies, for example, at altitudes of 1.5–2.8 km on July 30, 2015 [see Fig. 5(a)] and 0.5–1.9 km on August 1, 2015 [see Fig. 5(b)], shall be attributed to geographical location differences for the two observations. It is concluded that the SRRL can yield reasonable WVMR results.

Combining the referencing N_2 Raman signals and applying the developed method, Fig. 6 presents the lidar-derived total backscatter spectra β^{T} (diamond) and decomposed backscatter

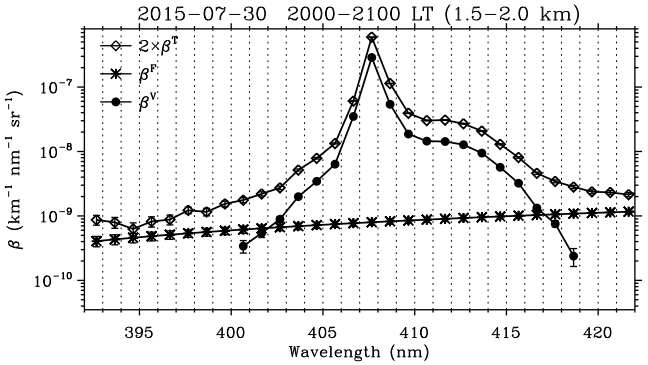


Fig. 6. Lidar-derived total backscatter spectra β^{T} (diamond) and decomposed backscatter spectral components of fluorescence β^{F} (asterisk) and water vapor β^{V} (solid dot) during 2000–2100 LT on July 30, 2015. The plotted β^{T} is multiplied by a factor of 2 for better discrimination. PC are accumulated in 1.5–2.0 km altitude range and error bars indicate the 1-σ uncertainties.

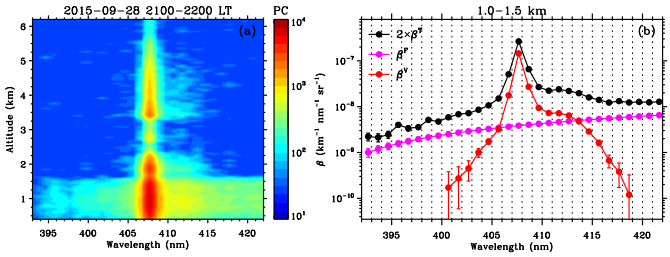


Fig. 7. (a) Altitude-resolved spectral signals measured by the SRRL during 2100–2200 LT on September 28, 2015. (b) Lidar-derived total backscatter spectra β^{T} (black) and decomposed backscatter spectral components of fluorescence β^{F} (magenta) and water vapor β^{V} (red). The plotted β^{T} is multiplied by a factor of 2 for better discrimination. Photons are accumulated in the altitude range of 1.0–1.5 km and error bars indicate the 1-σ uncertainties.

spectral components of fluorescence β^{F} (asterisk) and water vapor β^{V} (solid dot) during 2000–2100 LT on July 30, 2015. PC was accumulated in the altitude range of 1.5–2.0 km to generate the results. The plotted β^{T} is multiplied by a factor of 2 for better discrimination and error bars indicate the 1-σ uncertainties for the retrieved backscatter spectra results. For this clear summer case, the values of obtained fluorescence backscatter spectra β^{F} spanned a range of 4.07×10^{-10} to 1.22×10^{-9} km^{-1} nm^{-1} sr^{-1}, while the water vapor Q-branch Raman backscatter intensity was dominating: the value of β^{V}_{16} (2.88×10^{-7} km^{-1} nm^{-1} sr^{-1}) was about 360 times larger than that of β^{F}_{16} (7.98×10^{-10} km^{-1} nm^{-1} sr^{-1}). This indicates that the aerosol fluorescence contamination to water vapor Q-branch Raman signal is reasonably negligible, so conventional water vapor Raman lidars work well in clear areas.

B. Fluorescence Measurement Case

Fig. 7 shows the SRRL measurement results during 2100–2200 LT on September 28, 2015. Generally, the atmosphere was moderately clear without clouds, but aerosols were present in the lower atmosphere as inferred by the co-located 532-nm polarization lidar [43] at the Atmospheric Observatory. Fig. 7(a) plots the altitude-resolved spectral signals. It is seen that except for the prominent water vapor

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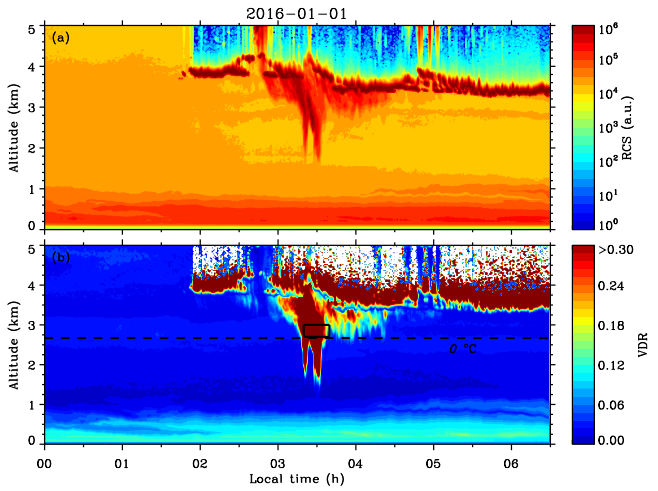


Fig. 8. (a) Range-corrected signals (RCS) and (b) volume depolarization ratio (VDR) obtained by the co-located polarization lidar during 0000–0630 LT on January 1, 2016. The temporal and spatial resolutions are 1 min and 30 m, respectively. Horizontal dashed line stands for the 0 °C level inferred by temperature data of radiosonde launched at 0800 LT. Black rectangle indicates the time and altitude spans for studying the mixed-phase water virga.

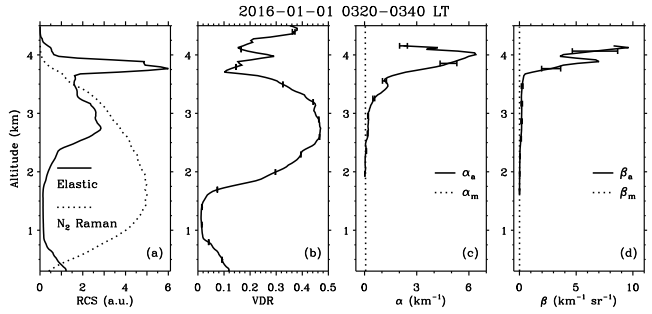


Fig. 9. Profiles of (a) range-corrected N_2 Raman signals (dotted) from the SRRL and elastic signals (solid) form the polarization lidar; (b) VDR by the polarization lidar; (c) particle extinction coefficient α_a retrieved by using the N_2 Raman signals; and (d) backscatter coefficient β_a by the two lidars (solid); molecular extinction coefficient α_m and backscatter coefficient β_m (dotted) are added for reference. Data during 0320–0340 LT on January 1, 2016, were used to generate the results with a vertical resolution of 30 m. Error bars indicate the 1-σ uncertainties.

Q-branch Raman peak signals in the 16th channel, the spectral signals on both sides were also discernible in the altitudes below ~1.5 km. While different from those shown in Fig. 3(a), at an altitude below 1.5 km, the intensities of spectral signals on both sides (especially on the longer-wavelength side) of the peak generally showed a gentle decreasing trend. Using PC data accumulated in the 1.0–1.5 km altitude range, Fig. 7(b) presents the lidar-derived total backscatter spectra β^T (black) and decomposed backscatter spectral components of fluorescence β^F (magenta) and water vapor β^V (red). As shown in Fig. 7(b), the values of fluorescence backscatter spectra β^F ranged from 1.00×10^{-9} to 6.88×10^{-9} km^{-1} nm^{-1} sr^{-1}, at least 2.4 times larger than those shown in Fig. 6. The reason for increased fluorescence backscatter intensities might be due to gathered fluorescent aerosols with a larger concentration in the boundary layer during the lidar observational period. Besides, it was checked that the water vapor Q-branch Raman backscatter intensity β_{16}^V (1.45×10^{-7} km^{-1} nm^{-1} sr^{-1}) was only about 37 times larger than that of β_{16}^F (3.85×10^{-9} km^{-1} nm^{-1} sr^{-1}), which should lead to about 2.7% larger WVMR results for conventional water vapor Raman lidars. Actually, fluorescence was frequently observed by the SRRL, especially in the lower troposphere (atmospheric boundary layer; see also Fig. 11). This case indicates that SRRL can also perform aerosol fluorescence measurements and distinguish water Raman signals from the contaminating aerosol fluorescence.

C. Mixed-Phase Water Virga Measurement Case

Fig. 8 presents the measurement results of range-corrected signals (RCS) and volume depolarization ratio (VDR) by the co-located polarization lidar during 0000–0630 LT on January 1, 2016. The temporal and spatial resolutions are 1 min and 30 m, respectively, for the lidar. As shown by Fig. 8(a), a stratiform cloud was present with a clear-cut cloud base altitude decreasing from ~3.7 km at 0150 LT to ~3.2 km at 0630 LT. At around 0300 LT, another layer with strong RCS [see Fig. 8(a)] and large VDR [see Fig. 8(b)] appeared below the cloud base until ~0420 LT. As an example and for detail, Fig. 9 provided the corresponding profile measurement results of RCS, VDR, particle extinction coefficient α_a, and backscatter coefficient β_a during 0320–0340 LT. As shown in Fig. 9, unlike the cloud above, the layer, especially at altitudes of 2.5–3.3 km, exhibited larger VDR with values of >0.4 [see Fig. 9(b)], but smaller α_a [see Fig. 9(c); solid] and β_a [see Fig. 9(d); solid]. This indicated that the layer contained condensed hydrometeors (ice crystals/water droplets) that fell out of the stratiform cloud above. During 0320–0340 LT, the cloud base and top altitudes were ~3.6 and ~4.0 km [see Fig. 9(a)], respectively, with corresponding temperatures of ~−5.6 °C and ~−6.7 °C according to temperature data of radiosonde launched at 0800 LT from the Wuhan Weather Station. Rangno and Hobbs [56] suggested that for slightly supercooled stratiform clouds with cloud top temperature warmer than −4 °C, precipitation of drizzle droplets might possibly occur [57], [58]; while when cloud top temperature decreased to be −4~−10 °C, both drizzle droplets and ice crystals could form and fall out from the cloud. Hence, we cannot conclude that this layer contained only ice-phase water (ice crystals) even though the overall VDR was large. Note that the falling condensed hydrometeors did not reach the ground, but gradually disappeared in their falling path. Between ~0318 and 0335 LT, the falling condensed hydrometeors crossed the 0 °C level [see Fig. 8(b); horizontal dashed line].

Luckily, the SRRL also operated on this night and recorded the corresponding spectral signals. Here, we designate it as a "mixed-phase water virga" measurement case. For improving the SNRs of the SRRL echoes, the photons in the altitude range of 2.7–3.0 km during 0320–0340 LT, as indicated by the black rectangle in Fig. 8(b), were accumulated to generate the subsequent result. Fig. 10(a) plots the altitude-resolved spectral signals measured by the SRRL during 0320–0340 LT on January 1, 2016. It is clearly seen that there were intensified spectral signals between 2.5 and 3.3 km.

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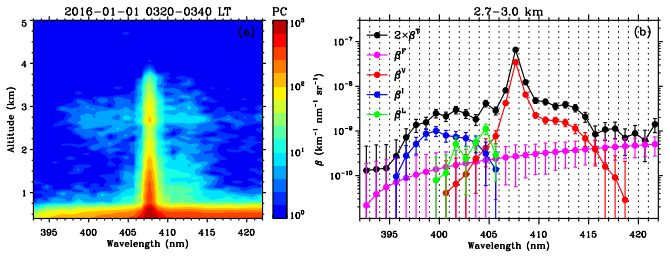


Fig. 10. (a) Altitude-resolved spectral signals measured by the SRRL during 0320–0340 LT on January 1, 2016. (b) Lidar-derived total backscatter spectra β^{T} (black) and decomposed backscatter spectra components of fluorescence β^{F} (magenta), water vapor β^{V} (red), ice water β^{I} (blue), and possible liquid water β^{L} (green). The plotted β^{T} is multiplied by a factor of 2 for better discrimination. Photons are accumulated at altitudes of 2.7–3.0 km and error bars indicate the 1-σ uncertainties. Note ice water Raman spectra from Slusher and Derr [15] are used for decomposition of the condensed water Raman spectra.

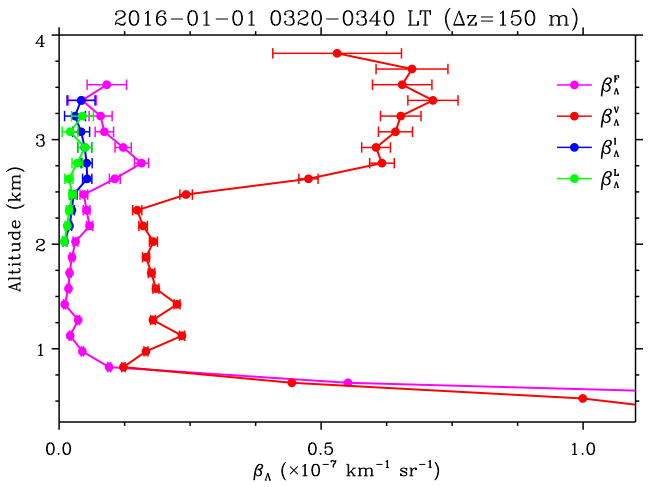


Fig. 11. Profiles of integrated total volume backscattering coefficients of fluorescence β_Λ^{F} (magenta), water vapor β_Λ^{V} (red), ice water β_Λ^{I} (blue), and possible liquid water β_Λ^{L} (green) measured by the SRRL during 0320–0340 LT on January 1, 2016. Vertical resolution is 150 m and error bars indicate the 1-σ uncertainties.

Accordingly, Fig. 10(b) provides the backscatter spectral measurement results. After retrieval of total backscatter spectra β^{T} (black) and subtraction of fluorescence backscatter spectra β^{F} (magenta) and water vapor Raman backscatter spectra β^{V} (red), the condensed water Raman backscatter spectra β^{C} (not shown) were first decomposed using the theoretical ice water Raman spectra from Slusher and Derr [15] to retrieve and represent ice water Raman backscatter spectra β^{I} (blue), then the residual backscatter spectra were plotted in green. It was found that the condensed water Raman backscatter spectra were quite similar in shape to the theoretical ice water Raman spectra from Slusher and Derr [15] in the 395–401 nm range for this mixed-phase water virga case. This is consistent with the fact that this virga contained ice crystals. As to the residual backscatter spectra (green), they mainly fell into the 399–406 nm spectral range and had spectra shape roughly similar to that of bulk liquid water Raman spectra [15], [50], even if the 1-σ uncertainties were relatively large due to low SNRs. It is inferred that accompanying the falling ice crystals, there was also less amount of liquid water (droplets) present in the path. So, for this lidar-probed mixed-phase water virga, ice crystals dominated over water droplets. Consequently, the residual backscatter spectra might be attributed to liquid water Raman backscatter spectra here. Note that intensities of fluorescence backscatter spectra β^{F} (ranged from 2.17×10^{-11} to 5.40×10^{-10} km^{-1} nm^{-1} sr^{-1}) were generally comparable to those of condensed ice/liquid water for this mixed-phase water virga measurement case, indicating that separation and elimination of fluorescence backscatter serve as a prerequisite for accurate measurements of condensed liquid/ice water content for water Raman lidars.

Fig. 11 presents the vertical distributions of integrated total volume backscattering coefficients of fluorescence β_Λ^{F} (magenta), water vapor β_Λ^{V} (red), ice water β_Λ^{I} (blue), and possibly liquid water β_Λ^{L} (green) measured by the SRRL during 0320–0340 LT on January 1, 2016 (vertical resolution set to be 150 m). It is seen in Fig. 11 that at altitudes of 2.0–3.5 km, the β_Λ^{F} value grew to be $\sim(3.13–15.72) \times 10^{-9}$ km^{-1} sr^{-1}, but it can be very small ($<2 \times 10^{-9}$ km^{-1} sr^{-1}) at clearer areas (e.g., $\sim$1.4 km; see Fig. 8); this indicates that the lidar-measured fluorescence due to receiving optics does not contribute but shall mainly be from fluorescent aerosols. At altitudes of 2.0–3.5 km, both β_Λ^{I} and β_Λ^{L} were discernible with smaller amplitudes, hinting at the possible coexistence of ice crystals and water droplets. Note for this mixed-phase water virga measurement case, the lidar-retrieved β_Λ^{I} was generally larger than β_Λ^{L} and the overall VDR was large [>0.3; Fig. 9(b)]. This alerts that a polarization lidar alone might fail to infer the presence of liquid water when its backscatter intensity was less than that of ice water. By contrast, an SRRL easily achieves to distinguish all three phase-states water; moreover, the lidar-measured β_Λ^{I} and β_Λ^{L} actually reflect the ice and liquid water content according to (19). In conclusion, the developed spectral decomposition method is capable of yielding backscatter spectra of fluorescent aerosol, water vapor, ice water, and liquid water successively, and the SRRL can perform simultaneous measurements of fluorescent aerosols and all three-phase water.

V. Summary and Outlook

An SRRL has been presented for simultaneous measurements of atmospheric three-phase water and fluorescent aerosols. The lidar transmitter emits 354.8-nm laser light, while the receiver uses a 450-mm telescope for collecting atmospheric backscatter. Both the N_2 Raman signals around 386.7 nm and spectral signals in the $\sim$393–424 nm range are registered by the SRRL. A DGP with a linear dispersion of 1.0 mm nm^{-1} performs dispersion of spectral signals in the $\sim$393–424 nm range, and a 32-channel detector records the dispersed signals with 1.0-nm resolution. A joint usage of bandpass filters and the DGP in the receiver yields a suppression of more than 15 orders of magnitude to the elastic signal around 354.8 nm, and suppression of more than six orders of magnitude to the O_2 and N_2 vi-rotational Raman signals around 375.4 and 386.7 nm, respectively. So the recorded signals in the spectral range of $\sim$393–424 nm are not

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contaminated by the elastic signals as well as the O_2 and N_2 Raman signals.

The lidar spectral range of ~393–424 nm is specially set. The ~395–418 nm Raman band of three-phase water is intended for three-phase water measurement. The spectral ranges of ~393–394 and ~419–424 nm are used for studying fluorescent aerosols. Moreover, by assuming that the aerosol fluorescence vary smoothly in the limited lidar spectral range, a linear fitting using spectral signals in these two spectral ranges is applied to estimate the whole fluorescence spectra in the ~393–424 nm range. Subsequently, a unique spectral decomposition method is developed for the retrieval of normalized spectral components of fluorescence and normalized Raman spectral components of water vapor, ice water, and liquid water successively from the defined total normalized spectra. The advantage of defining normalized spectra is that the normalized spectra depend on water Raman spectra shape rather than absolute water content. Finally, by referencing the N_2 Raman signals, the absolute volume backscattering coefficient spectra (backscatter spectra for short) of fluorescent aerosols and three-phase water are also retrieved.

Three typical kinds of measurement cases are presented to verify the proposed method and the lidar performance. For the clear-day case, the normalized water vapor Raman spectra are investigated based on the results of eight separate nights. The lidar-derived total-mean normalized water vapor Raman spectra are found to be nearly invariant in shape in the ~400–419 nm range. This feature enables the total-mean normalized water vapor Raman spectra to serve as the "background spectra reference" for the decomposition of three-phase water Raman spectra. Besides, the intensities of fluorescence backscatter spectra are usually small enough to be neglected for water vapor measurements in clear areas. For the fluorescence case, the decomposition method successfully distinguishes water vapor Raman signals from the contaminating aerosol fluorescence, even if the relative fluorescence backscatter intensities have intensified to influence the accurate measurements of water vapor. For the mixed-phase water virga case, it is found that the decomposed Raman backscatter spectra of condensed water are quite similar in shape to the theoretical ice water Raman spectra from Slusher and Derr [15] in the 395–401 nm range. Residual weak backscatter spectra intensities in the 399–406 nm range are regarded as due to liquid water, indicating the possible coexistence of less amount of liquid water in the falling water virga.

For the protection of the lidar system, the SRRL mainly works under fine weather conditions without precipitation. It is planned to equip a window system [53] above the telescope so that the SRRL can also operate despite light rain or snow. A more powerful laser with an average power of ~12–15 W at 354.8 nm is scheduled to be placed instead of the current 2-W laser to improve the lidar detection performance. Additional elastic detection channels (e.g., by extracting elastic return signals before BP1) with polarization measurement capability can be added to the current SRRL for better characterization of atmospheric particles. So far, we have recorded only one measurement case of ice-phase water by this SRRL, and more ice water measurement cases are needed to investigate the lidar-measured normalized ice water Raman spectra.

Currently, the SRRL has yielded integrated three-phase water Raman backscattering coefficients $\beta_\Lambda^{\mathrm{W}}$ (W stands for water vapor V, liquid water L, and ice water I). For water vapor, the molecular number density n^{V} (and further water vapor content) can also be directly obtained by combining β_{16}^{V} and the known effective Raman differential backscattering cross section σ^{V} for the 16th detection channel. However, for the case of cloud liquid/ice water, things become much more complicated as the integrated liquid/ice water Raman differential backscattering cross section $\sigma_\Lambda^{\mathrm{L,I}}$ from a swarm of water droplets/ice crystals with distinct sizes are distinct from those of bulk liquid/ice water measured in lab. Actually, an enhancement factor [33] must be considered due to the structural resonance of cloud particles; very recently, the excellent work by Reichardt *et al.* [59] has proposed the approach for determination of the enhancement factor as well as consequent retrieval of accurate and absolute liquid/ice water content of both clouds and precipitation. Besides, temperature profiles from concurrent and co-located rotational temperature Raman lidars [44], [45] can be combined to correct for the temperature dependence of the N_2 and water Raman spectra [54], [55]. A follow-up work shall focus on the topic of retrieving absolute and accurate atmospheric water vapor and cloud liquid/ice water content using the SRRL.

It is concluded that the developed spectral decomposition method can yield backscatter spectra of fluorescent aerosols, water vapor, ice water, and liquid water successfully. Besides, combining polarization and temperature Raman lidars (or measurement results), this kind of SRRL is believed to have provided the potential for simultaneous and accurate profiling of atmospheric three-phase water, especially condensed liquid/ice water in clouds, and open up new perspectives for studies of cloud–aerosol interaction.

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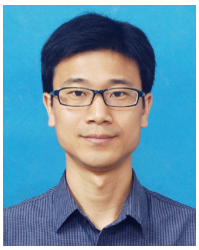
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